# DG INTERNS HUB — TECHNICAL NOTES

## TOPIC 05: SPLUNK ARCHITECTURE & DATA PIPELINE

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## 1. Core Architectural Components

- **Universal Forwarder (UF):** Lightweight endpoint agent designed exclusively to capture logs and forward them over TCP port 9997 without heavy processing.
- **Heavy Forwarder (HF):** Full Splunk instance that parses, filters, routes, or masks sensitive data before indexing.
- **Indexer (IN):** Receives raw logs, parses timestamps/sourcetypes, creates rawdata compressed chunks and index keys, and writes them to disk storage.
- **Search Head (SH):** User GUI front-end that translates SPL user inputs into distributed search jobs across Indexer clusters.
- **Deployment Server (DS):** Central management server pushing configuration files (`inputs.conf`, `outputs.conf`) to forwarder fleets.

## 2. The 4-Stage Data Pipeline

Data progresses through four distinct stages: `Input Stage (UF) → Parsing Stage (HF/Indexer splits lines & extracts time) → Indexing Stage (Writes to disk) → Search Stage (SH executes SPL).`

## 3. Index Bucket Lifecycle

Data stored in Splunk transitions through five age buckets: **Hot** (read/write) → **Warm** (read-only) → **Cold** (older read-only on secondary storage) → **Frozen** (archived or deleted) → **Thawed** (restored for audit investigations).

## 4. Practical SOC Scenario: Forwarder Ingestion Drop

> **Scenario: Firewall Logs Stop Ingesting into Splunk**
>
> **Symptom:** Dashboard shows zero events for `sourcetype=pan:traffic` for over 30 minutes.
>
> **Analyst Workflow:** Check Indexer deployment status in `_internal` index using: `index=_internal sourcetype=splunkd group=tcpin_connections | stats count by hostname, status.`
>
> **Root Cause:** Heavy Forwarder port 9997 connection blocked due to intermediate network firewall rule change.

## 5. Summary

Understanding Splunk's component separation ensures efficient query distribution, prevents log loss during network drops, and maintains high-availability SOC monitoring.